Harbor architecture review

Search team, second quarter

Alternatives considered (tundra)

- Streams replication logs so that a restart loses no acknowledged work
- Rejects hourly rollups after the leader confirms the lease

Operational runbook (lattice)

- Partitions pending writes without holding the global lock
- Archives expired sessions without holding the global lock
- Compacts replication logs once every replica reports healthy

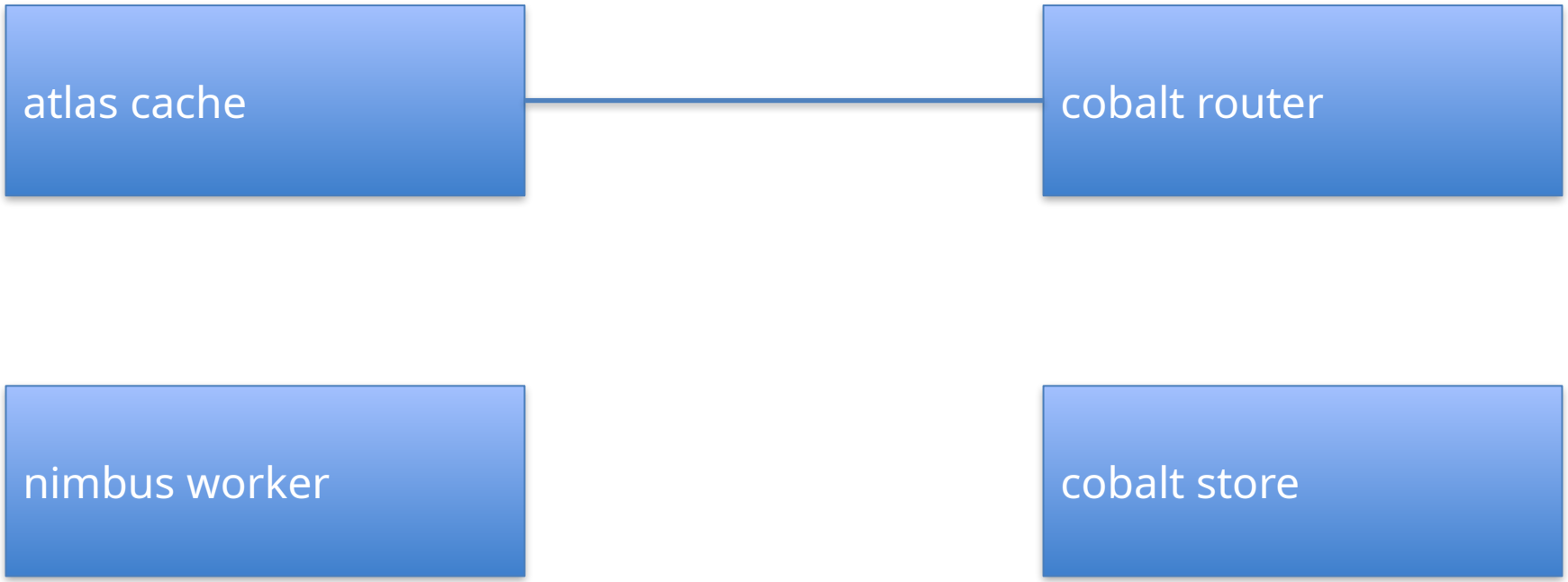
Proposed design (meridian)

- Archives delta snapshots so that a restart loses no acknowledged work
- Validates tenant quotas so that a restart loses no acknowledged work

Background (tundra)

Service	p50 ms	p99 ms	Error rate
harbor-0	30	2316	0.64%
kestrel-1	87	1809	0.20%
cobalt-2	2	279	0.46%

Read path (juniper)



Migration (ember)

- Signs hourly rollups without holding the global lock
- Streams index granules so a failed region drains within minutes
- Validates cold segments without holding the global lock
- Partitions pending writes under the tenant's own quota